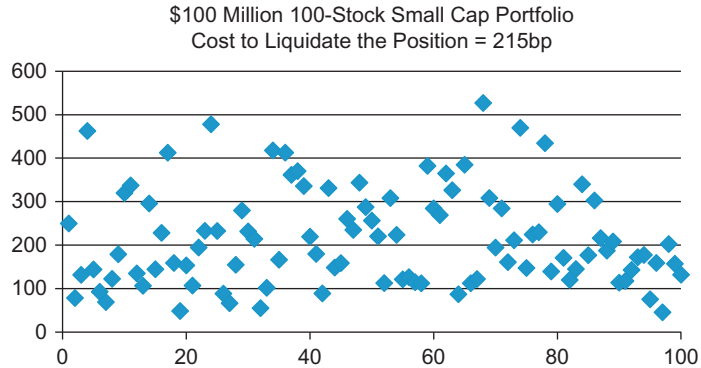




■ Figure 11.5 Cost to Liquidate Each Position in the 100 Stock Portfolio.

In a normal environment the market impact cost is:

$$MI_{Normal} = a_1 \cdot Size^{a_2} \cdot \sigma^{a_3} \cdot \left(\frac{Size}{1 + Size} \right)^{a_4} \quad (11.6)$$

In a stressed environment where volatility doubles and liquidity is halved the portfolio manager incorporates these expectations into the cost estimation model as follows:

$$MI_{Stressed} = a_1 \cdot Size^{a_2} \cdot (2 \cdot \sigma)^{a_3} \cdot \left(\frac{Size}{0.5 + Size} \right)^{a_4} \quad (11.7)$$

Therefore, in a stressed environment the cost premium is equal to:

$$Cost_{Premium} = 2^{a_3} \cdot \left(\frac{1}{0.5} \right)^{a_4} = 2^{a_3} \cdot 2^{a_4} = 2^{a_3 + a_4} \quad (11.8)$$

This cost premium for the full I-Star model can also be approximated from the simplified I-Star equation since the value of b_1 is often small. In our example, the increase in cost can be approximated with the above cost premium equation and our small cap market impact parameters as follows:

$$Cost_{Premium} = 2^{a_3 + a_4} = 2^{0.69 + 0.55} = 2^{1.24} = 2.36$$

Portfolio Management—Screening Techniques

Given the large potential variation in trading costs across stocks, managers have begun using many different techniques to screen and filter for